Instructions:

Step 1: Install Minikube (Windows)

Step 2: Verify Installation

minikube version
kubectl version --client

Step 3: Start Kubernetes Cluster

minikube start

Practical:

```
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> minikube version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> minikube start
* minikube v1.37.0 on Microsoft Windows 11 Home Single Language 10.0.26200.7462 Build 26200.7462
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.48 ...
* Updating the running docker "minikube" container ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical>
```

Instructions:

Step 4: Check Cluster Status

minikube status
kubectl cluster-info

Step 5: Check Nodes

kubectl get nodes

Step 6: Create Your First Pod

- Run an Nginx Pod
  kubectl run mypod --image=nginx

- Check Pod Status
  kubectl get pods

Practical:

```
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:49676
CoreDNS is running at https://127.0.0.1:49676/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   6d23h   v1.34.0
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl run mypod --image=nginx
pod/mypod created
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get pods
NAME                              READY   STATUS    RESTARTS        AGE
hello-minikube-7dd6984d9b-kwbtv   1/1     Running   1 (2m15s ago)   5m6s
hello-minikube-7dd6984d9b-r7lzw   1/1     Running   1 (2m15s ago)   5m6s
```

Instructions:

Step 7: Describe Pod (Very Useful)
kubectl describe pod mypod

Practical:

```
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl describe pod mypod
Name:             mypod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Wed, 14 Jan 2026 16:42:32 +0530
Labels:           run=mypod
Annotations:      <none>
Status:           Running
IP:               10.244.0.19
IPs:
  IP:  10.244.0.19
Containers:
  mypod:
    Container ID:   docker://509db1f3f5b870fda6856fe87c194ccab19b33ce85453d829f4a090cc9483c44
    Image:          nginx
    Image ID:       docker-pullable://nginx@sha256:e3a22a791adaf8f59de605d824f8d9cf0ac7c7c779d405bb0158144e57433246
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Wed, 14 Jan 2026 16:42:35 +0530
```

```
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-92mq4 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-92mq4:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
```

```
  PodScheduled                True
Volumes:
  kube-api-access-92mq4:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  3m9s  default-scheduler  Successfully assigned default/mypod to minikube
  Normal  Pulling    3m9s  kubelet            Pulling image "nginx"
  Normal  Pulled     3m6s  kubelet            Successfully pulled image "nginx" in 2.845s (2.845s including waiting). Image size: 160646317 byte
s.
  Normal  Created    3m6s  kubelet            Created container: mypod
  Normal  Started    3m6s  kubelet            Started container mypod
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical>
```

Step 8: Delete Pod

kubectl delete pod mypod

kubectl delete pods --all // or to delete all Pods

<u>Instructions:</u>

Step 9: Create Deployment (Real Way)

🧠 Deployment = Manages Pods automatically

kubectl create deployment web --image=nginx

Check Deployment

kubectl get deployments

kubectl get pods

Step 10: Scale Application

- Scale to 3 Pods

kubectl scale deployment web --replicas=3
kubectl get pods

Practical:

```
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl create deployment web --image=nginx
deployment.apps/web created
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get deployments
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube   2/2     2            2           29m
web              1/1     1            1           11s
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get pods
NAME                              READY   STATUS    RESTARTS        AGE
hello-minikube-7dd6984d9b-kwbtv   1/1     Running   1 (8m59s ago)   11m
hello-minikube-7dd6984d9b-r7lzw   1/1     Running   1 (8m59s ago)   11m
mypod                             1/1     Running   0               6m57s
web-64c966cf88-ndbvr              1/1     Running   0               20s
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl scale deployment web --replicas=3
deployment.apps/web scaled
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get pods
NAME                              READY   STATUS    RESTARTS        AGE
hello-minikube-7dd6984d9b-kwbtv   1/1     Running   1 (9m33s ago)   12m
hello-minikube-7dd6984d9b-r7lzw   1/1     Running   1 (9m33s ago)   12m
mypod                             1/1     Running   0               7m31s
web-64c966cf88-c5m48              1/1     Running   0               8s
web-64c966cf88-ndbvr              1/1     Running   0               54s
web-64c966cf88-nv6x6              1/1     Running   0               8s
```

Instructions:

Step 11: Scale Down
kubectl scale deployment web --replicas=1

Step 12: Self Healing Demo
kubectl delete pod <pod-name>

Step 13: Expose Application (Service)

- Create Service

kubectl expose deployment web --type=NodePort --port=80

Check Services
kubectl get services

Step 14: Access Application in Browser
minikube service web

Practical:

```
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl scale deployment web --replicas=1
deployment.apps/web scaled
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl expose deployment web --type=NodePort --port=80
service/web exposed
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical> kubectl get services
NAME                  TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
hello-minikube-service   NodePort    10.97.162.49    <none>        80:32230/TCP   32m
kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        7d
web                   NodePort    10.97.126.244   <none>        80:30710/TCP   9s
PS C:\Users\Satyam Raj\OneDrive\Desktop\Practice\Docker-Practical>
```

Step 15: Delete Resources (Cleanup)

- Delete Service
  kubectl delete service web
- Delete Deployment
  kubectl delete deployment web

🛑 Step 16: Stop Cluster
minikube stop

❌ Step 17: Delete Cluster Completely
minikube delete